



Lab#02

Department of Computer Science (7th Semester)

Course: DevOps

Name:

Zahra Bashir

SUBMITTED TO:

Sir Waqas Saleem

Registration Number

23-BCS-092(Section-A)

Lab Title:

Hands-On Git & Version Control Lab

Task 1: Create Private GitHub Repository

1. Create a new private repository named Lab2 on GitHub.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

 zahra225

Repository name *

DevOps_Lab2_Practice

✔ DevOps_Lab2_Practice is available.

Great repository names are short and memorable. How about [studious-enigma?](#)

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository.

 Private

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

No .gitignore

Task 2: Connect Repository via SSH

1. Generate a new SSH key using PowerShell:

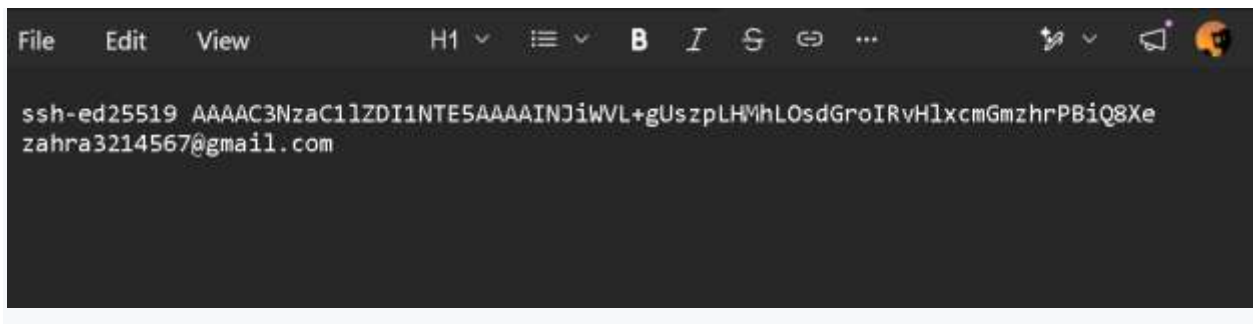
[illegible]

2. Add your SSH public key to GitHub (Settings > SSH and GPG keys).



3. Clone your Lab2 repo using SSH.

```
PS C:\WINDOWS\System32> cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGICYwYT0EqiZULScFPuuu/1h1wkPGjZftg8YBTy43e zahra3214567@gmail.com
PS C:\WINDOWS\System32> git clone git@github.com:zahra225/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
PS C:\WINDOWS\System32> cd DevOps_Lab2_Practice
PS C:\WINDOWS\System32\DevOps_Lab2_Practice> git remote -v
origin  git@github.com:zahra225/DevOps_Lab2_Practice.git (fetch)
origin  git@github.com:zahra225/DevOps_Lab2_Practice.git (push)
PS C:\WINDOWS\System32\DevOps_Lab2_Practice>
```



Task 3: Configure Git Username and Email

1. Set up your Git identity (this ensures all commits are linked to you):

```
PS C:\WINDOWS\System32\DevOps_Lab2_Practice> git config --global user.name Zahra Bashir
PS C:\WINDOWS\System32\DevOps_Lab2_Practice> git config --global user.email zahra3214567@gmail.com
PS C:\WINDOWS\System32\DevOps_Lab2_Practice>
```

2. Verify your configuration:

```

PS C:\WINDOWS\System32\DevOps_Lab2_Practice> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Zahra Bashir
user.email=zahra3214567@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=git@github.com:zahra225/DevOps_Lab2_Practice.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
PS C:\WINDOWS\System32\DevOps_Lab2_Practice> |

```

Task 4: Explore the .git Folder

```

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ ls -a .git
./ ../ HEAD config description hooks/ info/ objects/ refs/

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ |

```

Task 5: Local Repository Management

1. Delete the existing .git folder from your cloned repo using Git Bash:

```

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice
$ rm -rf .git

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice
$ git status
fatal: not a git repository (or any of the parent directories): .git

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice
$ |

```

2. Re-initialize the local git repository:

```
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice
$ git init
Initialized empty Git repository in C:/Users/GC/DevOps_Lab2_Practice/.git/

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    DevOps_Lab2_Practice/
    Labs/

nothing added to commit but untracked files present (use "git add" to track)

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (master)
$ |
```

3. Add a file named README.md and commit it:

```
GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice
$ git init
Initialized empty Git repository in C:/windows/System32/DevOps_Lab2_Practice/.git/

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (master)
$ echo "# Lab02 Git Practice" >> README.md

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (master)
$ git commit -m "Initial commit"
[master (root-commit) 9897065] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (master)
$ |
```

4. Connect your local repo to GitHub and push:

```
GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (master)
$ git remote add origin git@github.com:zahra225/DevOps_Lab2_Practice.git

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (master)
$ git branch -M main

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 239 bytes | 79.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:zahra225/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ |
```

Task 6: File Status & Staging

Create a new file notes.txt and write a note.

```

GC@Zahra MINGW64 /c:/windows/System32/DevOps_Lab2_Practice (main)
$ echo "Adding some test notes" >> notes.txt
GC@Zahra MINGW64 /c:/windows/System32/DevOps_Lab2_Practice (main)
$ ls
README.md  notes.txt
GC@Zahra MINGW64 /c:/windows/System32/DevOps_Lab2_Practice (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

nothing added to commit but untracked files present (use "git add" to track)
GC@Zahra MINGW64 /c:/windows/System32/DevOps_Lab2_Practice (main)

```

Check status:

```

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    DevOps_Lab2_Practice/
    Labs/
    notes.txt

nothing added to commit but untracked files present (use "git add" to track)
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ |

```

Stage and commit:

```

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git add note.txt
fatal: pathspec 'note.txt' did not match any files

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git commit -m "Update notes.txt"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   notes.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    DevOps_Lab2_Practice/
    Labs/

no changes added to commit (use "git add" and/or "git commit -a")

```

Edit notes.txt and repeat status/commit steps.

```

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git add note.txt
fatal: pathspec 'note.txt' did not match any files

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git commit -m "Update notes.txt"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.txt

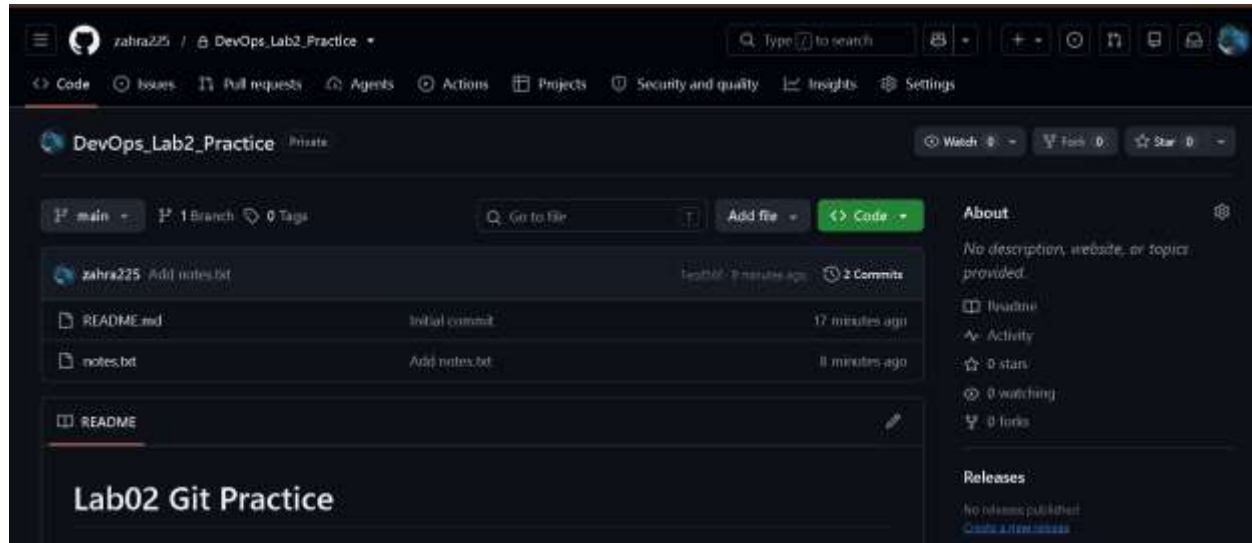
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        DevOps_Lab2_Practice/
        Labs/

no changes added to commit (use "git add" and/or "git commit -a")

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 304 bytes | 304.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:zahra225/DevOps_Lab2_Practice.git
   c013aeb..1ecd56f  main -> main

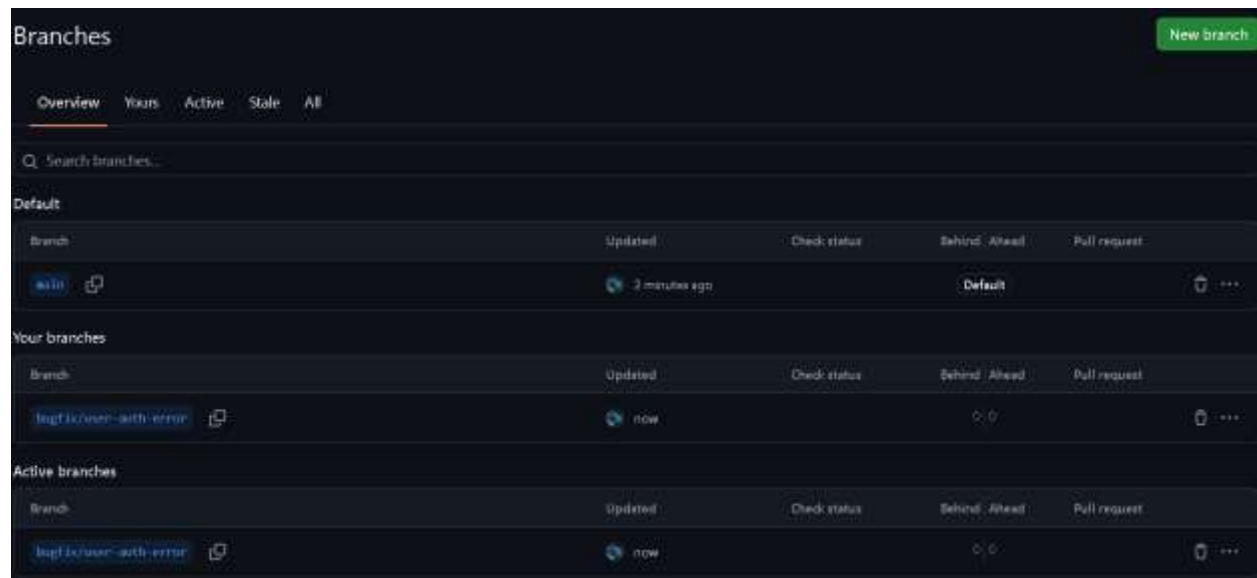
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ |

```

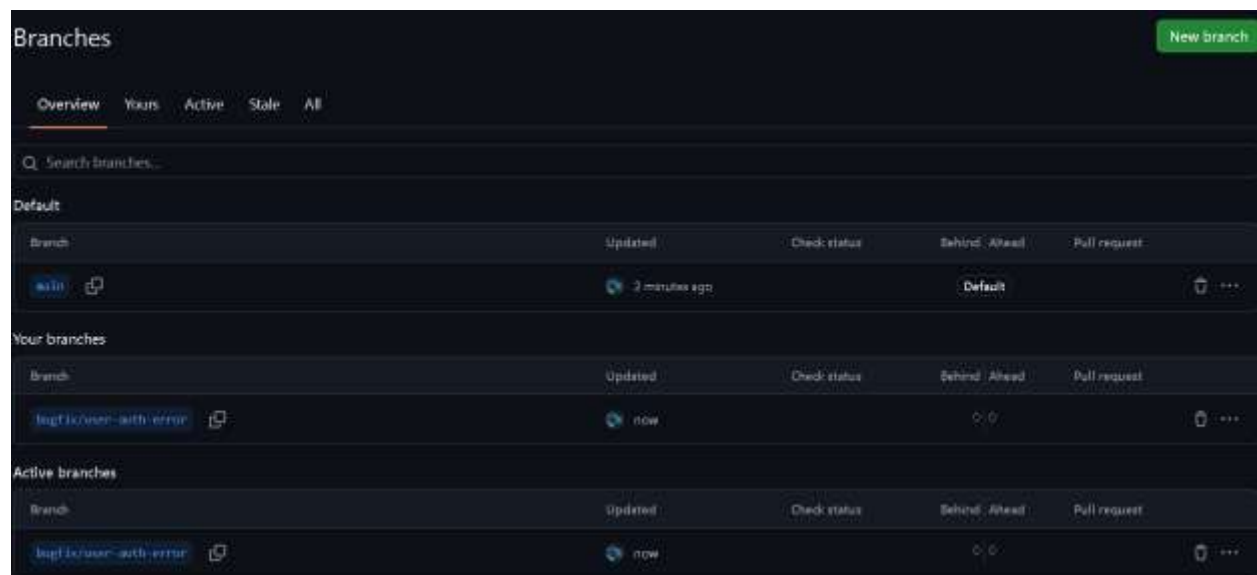


Task 7: Branch Creation Using GitHub GUI

1. On GitHub (web interface), create a branch named bugfix/user-auth-error.



2. Pull the branch to your local repository to sync.
3. git pull origin bugfix/user-auth-error



```

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git pull origin bugfix/user-auth-error
From github.com:zahra225/DevOps_Lab2_Practice
 * branch          bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]     bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git branch -a
* main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/main

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ |

```

Task 8: Branch Creation and Push Using Git Bash

Create a branch named feature/db-connection using Git Bash:

git checkout -b feature/db-connection

Push the branch to the remote repository:

```

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (feature/db-connection)
$ git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:   https://github.com/zahra225/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:zahra225/DevOps_Lab2_Practice.git
 * [new branch]     feature/db-connection -> feature/db-connection

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (feature/db-connection)
$ |

```

Task 9: Branching & Merging

1. Create and switch to a branch feature-1:

```

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (feature/db-connection)
$ |
GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git checkout -b feature-1
Switched to a new branch 'feature-1'

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (feature-1)
$ git branch
* feature-1
  feature/db-connection
  main

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (feature-1)
$ cat > main.py << EOF
def greet(name):
    return f'hello, {name}!'

print(greet("DevOps"))
EOF
$ git add main.py
$ git commit -m "Add main.py with greet function"
warning: in the working copy of 'main.py', LF will be replaced by CRLF the next time Git touches it
[feature-1] 8a70b13: Add main.py with greet function
1 file changed, 6 insertions(+), 0 deletions(-)
create mode 100644 main.py

```

2. Modify main.py (add a function) and commit.

```
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ echo "def hello():
    print('Hello from feature-1')" > main.py

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ cat main.py
def hello():
    print('Hello from feature-1')

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ |
```

```
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ git add main.py
warning: in the working copy of 'main.py', LF will be replaced by CRLF the next time Git touches it

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ git commit -m "Add new function to main.py"
[feature-1 6eda853] Add new function to main.py
1 file changed, 2 insertions(+)
create mode 100644 main.py

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ |
```

3. Switch back to main and merge:

```
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (feature-1)
$ git checkout main
M   notes.txt
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git merge feature-1
Updating 1ecd56f..6eda853
Fast-forward
 main.py | 2 ++
1 file changed, 2 insertions(+)
create mode 100644 main.py

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git log --oneline --graph --all
* 6eda853 (HEAD -> main, feature-1) Add new function to main.py
* 1ecd56f (origin/main, origin/feature/db-connection, origin/bugfix/user-auth-error, feature/db-connection) Add notes.txt
* c013aeb Initial commit

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ |
```

4. Push all branches:

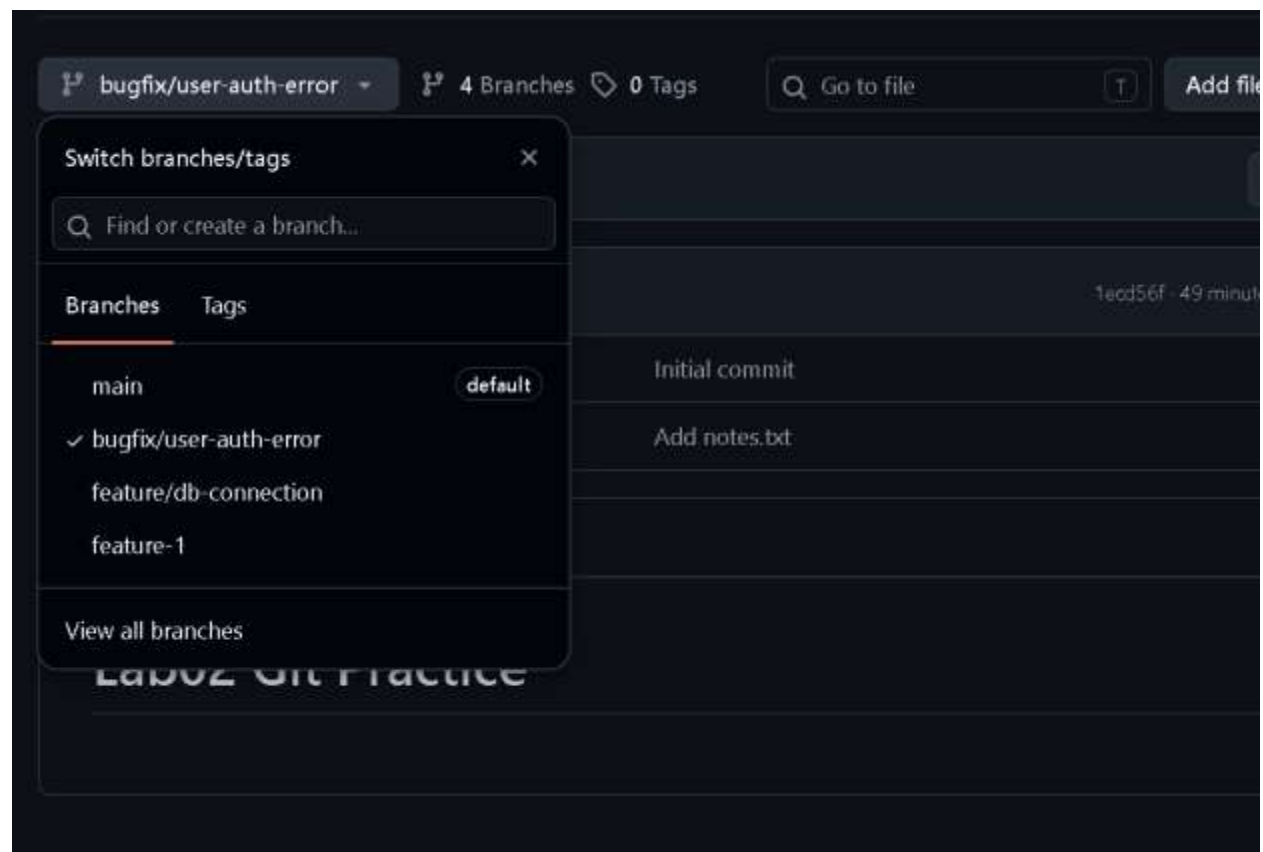
```

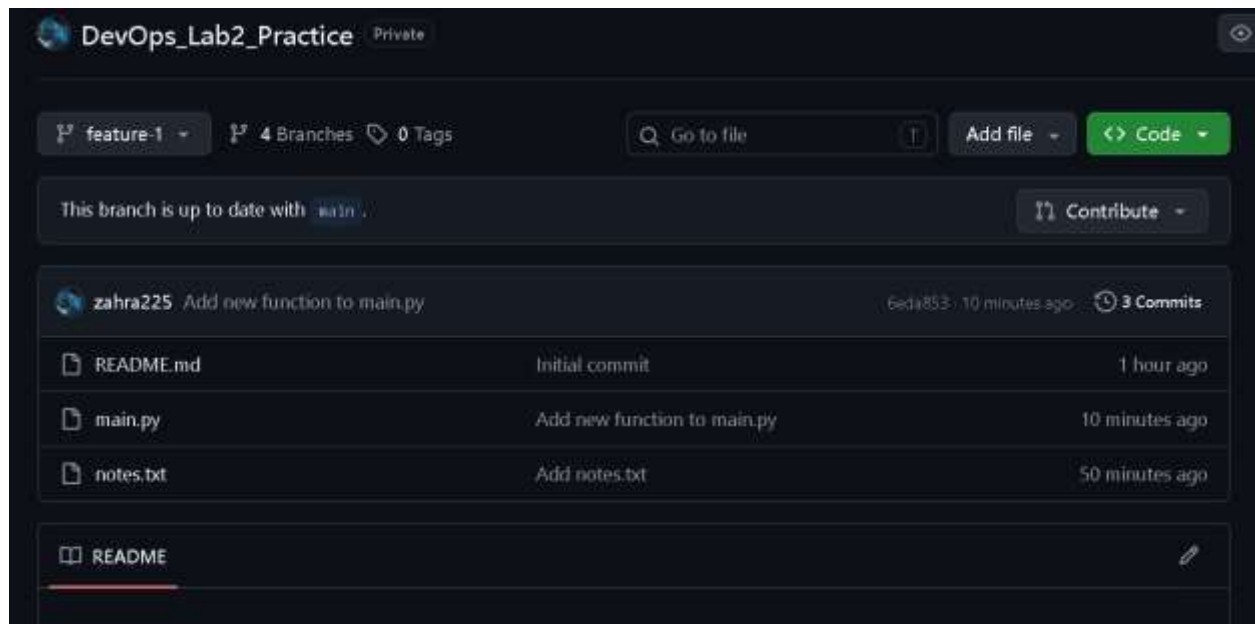
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 368 bytes | 368.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:zahra225/DevOps_Lab2_Practice.git
    1ecd56f..6eda853  main -> main

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:   https://github.com/zahra225/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:zahra225/DevOps_Lab2_Practice.git
   * [new branch]      feature-1 -> feature-1

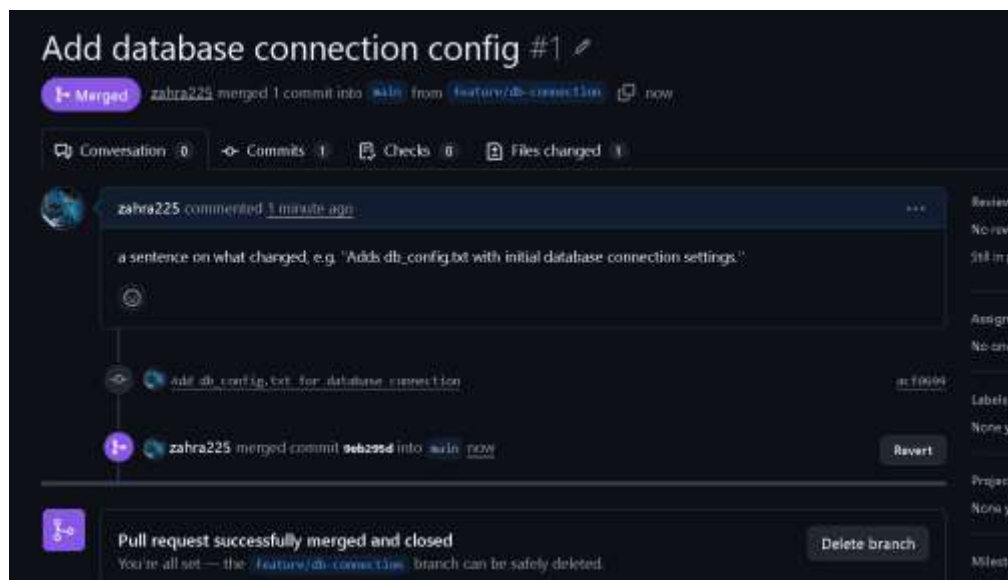
GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ |

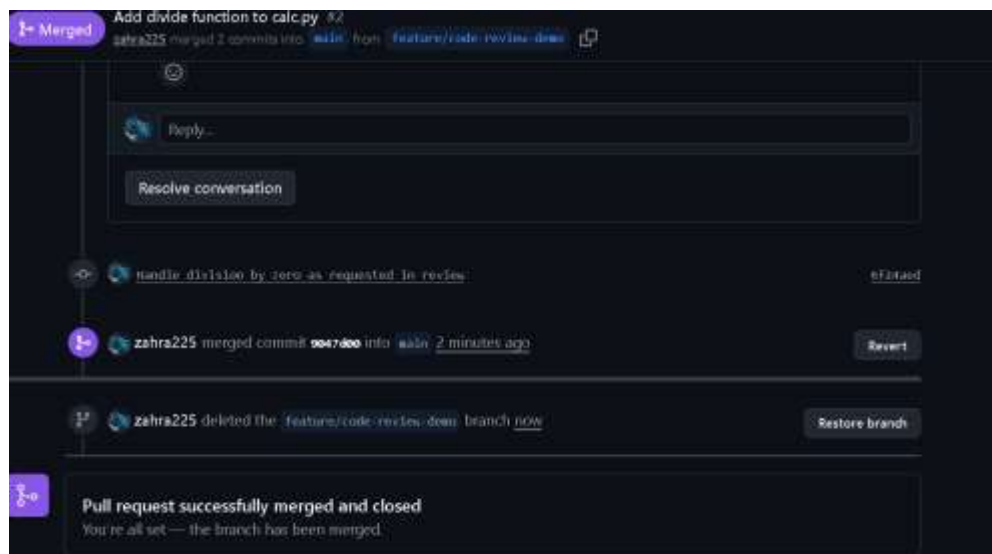
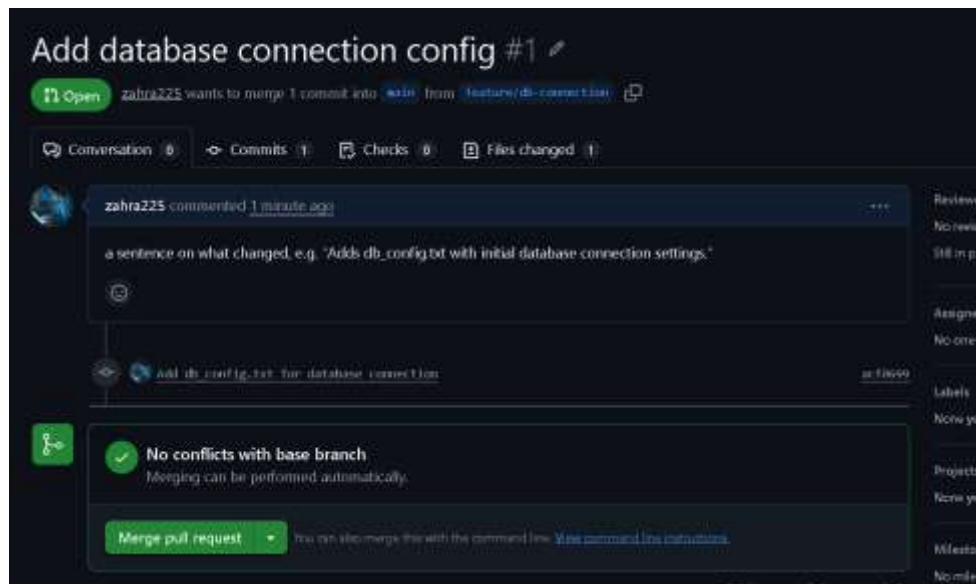
```





Task 10: Pull Request and Branch Review (GitHub GUI)





Task 11: Detailed Branch Strategy (Develop/Staging)


```

SC02Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git merge --no-ff staging -m "Release: merge staging into main"
Merge made by the 'ort' strategy.
  bugfix.txt | 1 +
  login.txt  | 1 +
  2 files changed, 2 insertions(+)
 create mode 100644 bugfix.txt
 create mode 100644 login.txt

SC02Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git push origin main
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 231 bytes | 231.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:zahra225/DevOps_Lab2_Practice.git
   9eb295d..98a2910  main -> main

SC02Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git log --one-line --graph --all
*   98a2910 (HEAD -> main, origin/main) Release: merge staging into main
|
| * facfcdf (origin/staging, staging) Merge develop into staging for testing
|
| * 58d0568 (origin/develop, develop) Merge bugfix/login-type into develop
|
| * 5974fa5 (origin/bugfix/login-type, bugfix/login-type) Fix typo in login page
|
| * 80171f3 Merge feature/login-page into develop
|
| * e75d90d (origin/feature/login-page, feature/login-page) Add login page
|
| * 9eb295d Merge pull request #1 from zahra225/feature/db-connection
|
| * acf0699 (origin/feature/db-connection, feature/db-connection) Add db_config.txt for database connection
|
| * 8a70633 (origin/feature-1, feature-1) Add main.py with greet function
|
|/
* 4ac1be0 (origin/bugfix/user-auth-error) Add notes.txt
* 9897065 Initial commit
SC02Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ |

```

Task 12: Code Review Workflow

The screenshot shows a GitHub pull request (PR) titled "Add divide function to calc.py #2". The PR is in the "Ready to merge" state, indicated by a green checkmark and the text "Ready to merge". The PR is created by user "zahra225" and is ready to be merged into the "main" branch. The PR description states: "adds a divide(a, b) function to calc.py." The PR includes a summary of changes and testing status. The summary shows that the file "calc.py" has been modified with the addition of the "divide()" function. The testing status is "Not yet tested." The PR is currently assigned to no one, and there are no reviews or labels. The PR is also not yet assigned to any projects or milestones.

Add divide function to calc.py #2

Ready to merge

Open | zahra225 wants to merge 1 commit into main from feature/code-review-225

Conversation | Commits | Checks | Files changed

Summary

adds a divide(a, b) function to calc.py.

Changes

- new file calc.py with divide()

Testing

Not yet tested.

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestones

Add divide function

zahra225 / DevOps_Lab2_Practice

Type / to search

CodeIssuesPull requestsActionsProjectsSecurity and qualityInsightsSettings

Add divide function to calc.py #2

Ready to mergeCode

Openzahra225 wants to merge 1 commit into main from feature/code-review-demo

ConversationCommitsChecksFiles changed

zahra225 commented 1 minute ago

Summary

Adds a divide(a, b) function to calc.py.

Changes

new file calc.py with divide()

Testing

not yet tested.

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

OpenAdd divide function to calc.py #2

zahra225 wants to merge 1 commit into main from feature/code-review-demo

rowaidajami975-glitch requested changes 2 minutes ago

View reviewed changes

rowaidajami975-glitch left a comment

Good start, but needs error handling before merge.

calc.py

def divide(a, b): return a / b

rowaidajami975-glitch 4 minutes ago

This will crash with ZeroDivisionError when b is 0. Please add a check and raise a clear error message.

Development

Successfully merging this pull request may cause these issues.

None yet

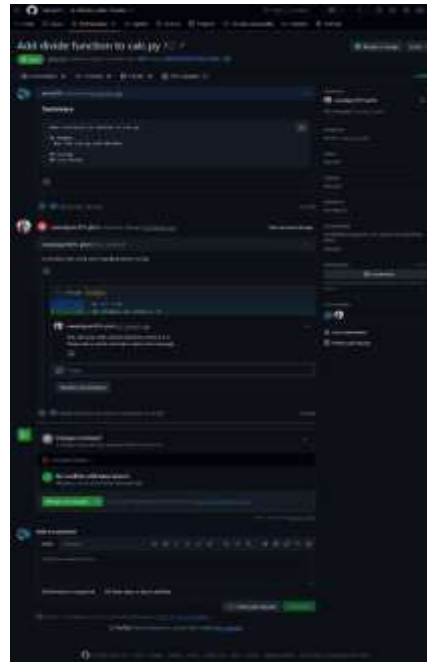
Notifications

Unsubscribe

You're receiving notifications because you commented.

2 participants

Lock conversation



Task 13: Branch Cleanup Best Practices

Default					
Branch	Updated	Check status	Behind / Ahead	Full request	
main	12 minutes ago		Default		...
Your branches					
Branch	Updated	Check status	Behind / Ahead	Full request	
feature/reactive-project	32 minutes ago		0 1	13	...
staging	yesterday		4 0		...
develop	yesterday		0 0		...
bugfix/login-typos	Deleted now		0 0		
feature/login-page	yesterday		0 0		...
View more branches					

```

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git branch -r --merged main
origin/HEAD -> origin/main
origin/bugfix/login-typo
origin/bugfix/user-auth-error
origin/develop
origin/feature-1
origin/feature/login-page
origin/main
origin/staging

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git pull origin main
From github.com:zahra225/DevOps_Lab2_Practice
* branch          main          -> FETCH_HEAD
Already up to date.

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git push origin --delete feature/login-page
To github.com:zahra225/DevOps_Lab2_Practice.git
- [deleted]          feature/login-page

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git fetch --prune
From github.com:zahra225/DevOps_Lab2_Practice
- [deleted]          (none)        -> origin/bugfix/login-typo

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ git branch -r
origin/HEAD -> origin/main
origin/bugfix/user-auth-error
origin/develop
origin/feature-1
origin/feature/practice-reject
origin/main
origin/staging

GC@Zahra MINGW64 /c/windows/System32/DevOps_Lab2_Practice (main)
$ !

```

```

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git branch -a
bugfix/login
develop
feature-1
feature/code-review
feature/db-connection
feature/payment
feature/rejected-pr
feature/review-demo
* main
staging
remotes/origin/bugfix/login
remotes/origin/bugfix/user-auth-error
remotes/origin/develop
remotes/origin/feature-1
remotes/origin/feature/code-review
remotes/origin/feature/db-connection
remotes/origin/main
remotes/origin/staging

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git branch -d feature/payment
Deleted branch feature/payment (was 37757e5).

GC@Zahra MINGW64 ~/DevOps_Lab2_Practice (main)
$ git branch
bugfix/login
develop
feature-1
feature/code-review
feature/db-connection
feature/rejected-pr

```